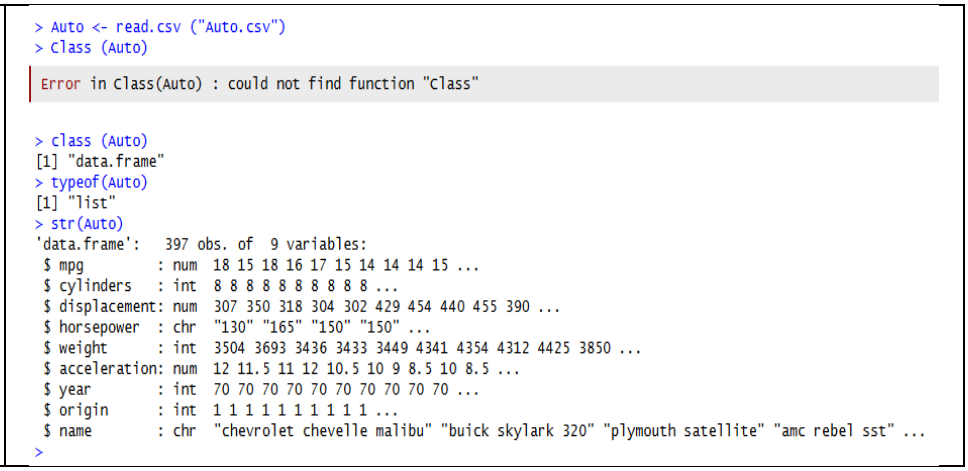
Christopher Horyczun

February 8, 2026

Cs 2020 (Friday)

Assignment 1

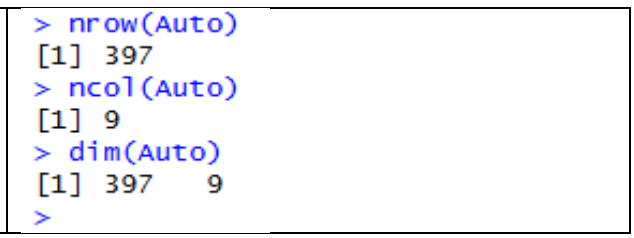
Question 1:

The Auto dataset was loaded into R as a data frame. The str() function is showing us the number of observations, variables, and data types for each column.	

```
> Auto <- read.csv ("Auto.csv")
> Class (Auto)
Error in Class(Auto) : could not find function "Class"

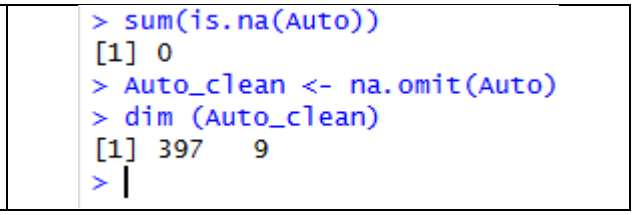
> class (Auto)
[1] "data.frame"
> typeof(Auto)
[1] "list"
> str(Auto)
'data.frame':   397 obs. of  9 variables:
 $ mpg         : num  18 15 18 16 17 15 14 14 14 15 ...
 $ cylinders   : int  8 8 8 8 8 8 8 8 8 8 ...
 $ displacement: num  307 350 318 304 302 429 454 440 455 390 ...
 $ horsepower  : chr  "130" "165" "150" "150" ...
 $ weight      : int  3504 3693 3436 3433 3449 4341 4354 4312 4425 3850 ...
 $ acceleration: num  12 11.5 11 12 10.5 10 9 8.5 10 8.5 ...
 $ year        : int  70 70 70 70 70 70 70 70 70 70 ...
 $ origin      : int  1 1 1 1 1 1 1 1 1 1 ...
 $ name        : chr  "chevrolet chevelle malibu" "buick skylark 320" "plymouth satellite" "amc rebel sst" ...
>
```

Question 2:

The dataset contains 397 obs. of 9 variables. The nrow() and ncol() functions were used to determine the size of this dataset.	

```
> nrow(Auto)
[1] 397
> ncol(Auto)
[1] 9
> dim(Auto)
[1] 397   9
>
```

Question 3:

The dataset showed to have no missing values. After removing rows with missing values using na.omit(), the cleaned dataset contains 397 observations and 9 variables.	

```
> sum(is.na(Auto))
[1] 0
> Auto_clean <- na.omit(Auto)
> dim (Auto_clean)
[1] 397   9
>
```

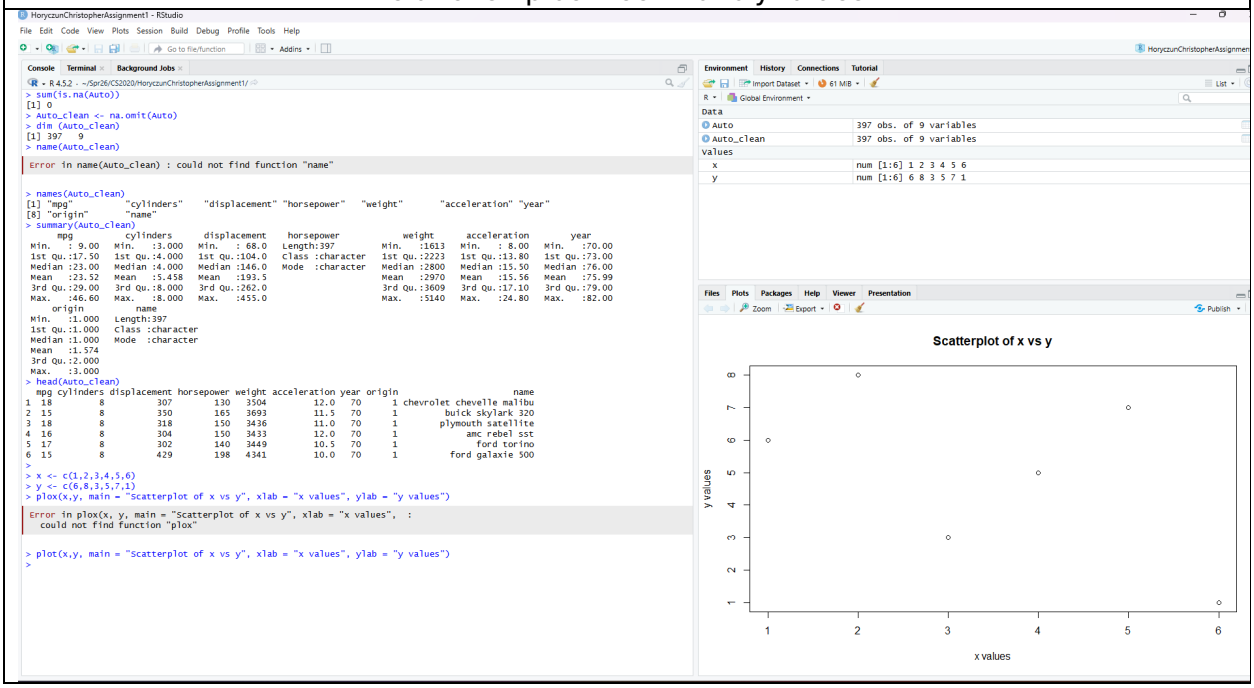
Question 4:

The names(), summary(), and head() functions were used to explore the dataset. Cylinders, origin, and name are categorical. Mpg, horsepower, and weight are continuous

```
> names(Auto_clean)
[1] "mpg"          "cylinders"    "displacement" "horsepower"   "weight"       "acceleration" "year"
[8] "origin"       "name"
> summary(Auto_clean)
      mpg          cylinders      displacement    horsepower          weight      acceleration        year
 Min.   : 9.00   Min.   :3.000   Min.   : 68.0   Length:397         Min.   :1613   Min.   : 8.00   Min.   :70.00
 1st Qu.:17.50   1st Qu.:4.000   1st Qu.:104.0   Class :character   1st Qu.:2223   1st Qu.:13.80   1st Qu.:73.00
 Median :23.00   Median :4.000   Median :146.0   Mode  :character   Median :2800   Median :15.50   Median :76.00
 Mean   :23.52   Mean   :5.458   Mean   :193.5                      Mean   :2970   Mean   :15.56   Mean   :75.99
 3rd Qu.:29.00   3rd Qu.:8.000   3rd Qu.:262.0                      3rd Qu.:3609   3rd Qu.:17.10   3rd Qu.:79.00
 Max.   :46.60   Max.   :8.000   Max.   :455.0                      Max.   :5140   Max.   :24.80   Max.   :82.00
     origin          name
 Min.   :1.000   Length:397
 1st Qu.:1.000   Class :character
 Median :1.000   Mode  :character
 Mean   :1.574
 3rd Qu.:2.000
 Max.   :3.000
> head(Auto_clean)
  mpg cylinders displacement horsepower weight acceleration year origin                  name
1  18         8          307        130   3504         12.0   70      1 chevrolet chevelle malibu
2  15         8          350        165   3693         11.5   70      1         buick skylark 320
3  18         8          318        150   3436         11.0   70      1        plymouth satellite
4  16         8          304        150   3433         12.0   70      1            amc rebel sst
5  17         8          302        140   3449         10.5   70      1              ford torino
6  15         8          429        198   4341         10.0   70      1         ford galaxie 500
>
```

Question 5:

Two numeric vectors were created and plotted using a scatterplot. The plot shows the relationship between x and y values.

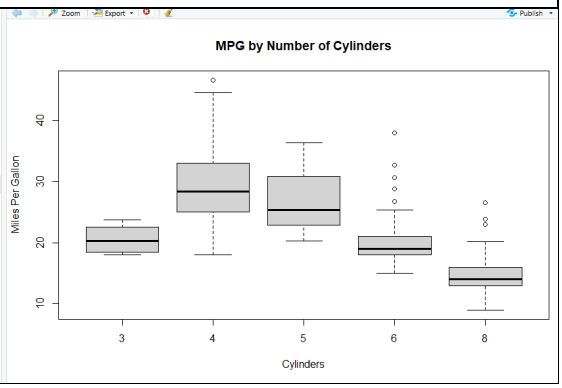


Question 6:

I converted the cylinders variable into a factor so that R would treat it as a categorical value instead of a number. After doing that, I created a boxplot to compare MPG across different cylinder counts, which shows that cars with fewer cylinders generally get better gas mileage.

```
> # Create numeric vector x
> x <- c(1, 2, 3, 4, 5, 6)
>
> # Create numeric vector y
> y <- c(6, 8, 3, 5, 7, 1)
>
> # Create a scatterplot of x vs y
> plot(x, y,
+      main = "Scatterplot of x vs y",
+      xlab = "x values",
+      ylab = "y values",
+      pch = 19)
>
> Auto_clean$cylinders <- as.factor(Auto_clean$cylinders)
> boxplot(mpg~ cylinders, date=Auto_clean, main = "MPG by number of cylinders",xlab = "Cylinders", ylab = "Miles per gallon")
Error in eval(predvars, data, env) : object 'mpg' not found

> boxplot(mpg ~ cylinders, data = Auto_clean, main = "MPG by Number of Cylinders", xlab = "Cylinders", ylab = "Miles Per Gallo
n")
>
>
```



Question 7:

At first, horsepower was not stored as a numeric variable, so it had to be converted before doing calculations. After converting it, I created a new variable that represents horsepower per unit of weight to compare car performance more easily.

```
> Auto_clean$hp_per_wt <- Auto_clean$horsepower / Auto_clean$weight
Error in Auto_clean$horsepower/Auto_clean$weight :
  non-numeric argument to binary operator

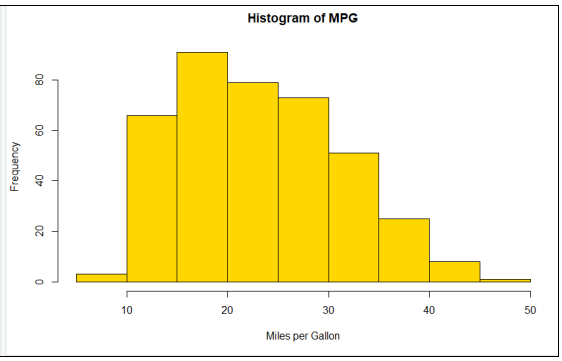
> Auto_clean$horsepower <- as.numeric(Auto_clean$horsepower)
Warning message:
NAs introduced by coercion

> Auto_clean$hp_per_wt <- Auto_clean$horsepower / Auto_clean$weight
> head(Auto_clean)
  mpg cylinders displacement horsepower weight acceleration year origin                       name  hp_per_wt
1  18         8          307        130   3504         12.0   70      1 chevrolet chevelle malibu 0.03710046
2  15         8          350        165   3693         11.5   70      1         buick skylark 320 0.04467912
3  18         8          318        150   3436         11.0   70      1        plymouth satellite 0.04365541
4  16         8          304        150   3433         12.0   70      1             amc rebel sst 0.04369356
5  17         8          302        140   3449         10.5   70      1               ford torino 0.04059148
6  15         8          429        198   4341         10.0   70      1          ford galaxie 500 0.04561161
>
```

Question 8:

I created a histogram of MPG using 12 bins to visualize how fuel efficiency is distributed across the dataset. Although 12 bins were specified, some bins are not visible because there are no cars in those MPG ranges. Overall, the histogram shows that most cars fall within the middle range of MPG values.

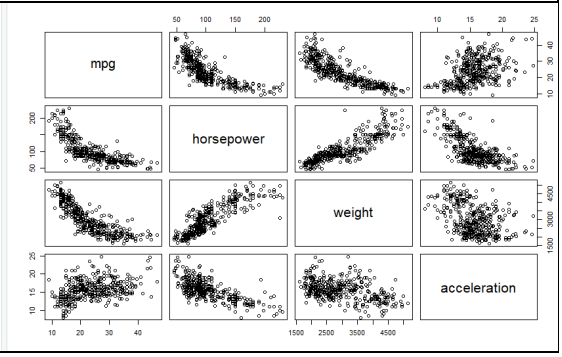
```
> hist(Auto_clean$mpg, main = "Histogram of MPG", xlab = "Miles per Gallon", col = "gold", breaks = 12)
>
```



Question 9:

I used a scatterplot matrix to compare MPG, horsepower, weight, and acceleration at the same time. The plots show a strong negative relationship between MPG and weight, meaning heavier cars tend to get worse gas mileage.

```
> hist(Auto_clean$mpg, main = "Histogram of MPG", xlab = "Miles per Gallon", col = "gold", breaks = 12)
> pairs(Auto_clean[ c ("mpg", "horsepower", "weight", "acceleration")])
>
```



Question 10:

Following the same idea as question 6, I built a boxplot; it focused on grouping mpg by cylinders. The plot shows that cars with fewer cylinders generally have higher MPG, while cars with more cylinders tend to have lower fuel efficiency.

```
> hist(Auto_clean$mpg, main = "Histogram of MPG", xlab = "Miles per Gallon", col = "gold", breaks = 12)
> pairs(Auto_clean[ c ("mpg", "horsepower", "weight", "acceleration")])
> Auto_clean$cylinders <- as.factor(Auto_clean$cylinders)
> boxplot(mpg = cylinders,data = Auto_clean
+         data = Auto_clean,

Error: unexpected symbol in:
"boxplot(mpg = cylinders,data = Auto_clean
        data"

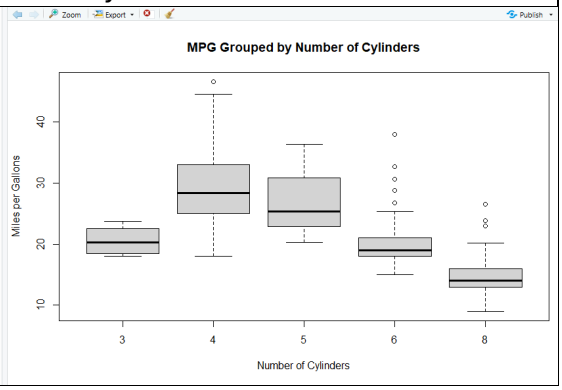
> boxplot(mpg = cylinders,data = Auto_clean
+ boxplot(mpg = cylinders,data = Auto_clean

Error: unexpected symbol in:
"boxplot(mpg = cylinders,data = Auto_clean
boxplot"

> boxplot(mpg = cylinders,
+
+
+
+
+
+
+ g)

Error: object 'g' not found

> boxplot(mpg ~ cylinders,
+ data = Auto_clean,
+ main = "MPG Grouped by Number of Cylinders",
+ xlab = "Number of Cylinders",
+ ylab = "Miles per Gallons")
>
```

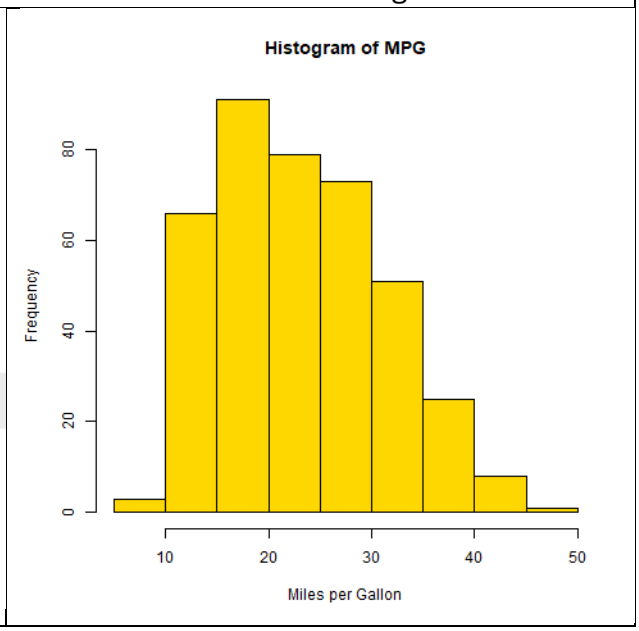


Question 11:

I saved the MPG histogram as a PNG file using the png() function. The plot was created while the file was open, and dev.off() was used to close and save the image.

```
> # Open a PNG file to save the plot
>
> png ("mpg_ histogram.png")
> #create the histogram of mpg
> hist(Auto_clean$mpg,)
> hist(Auto_clean$mpg,
+ main = "Histogram of MPG",
+ xlab = "Miles per Gallon",
+ breaks = 12,
+ col = "gold")
> $close the file and save the plot
Error: unexpected '$' in "$"

> #close the file and save the plot
> dev.off()
RStudioGD
        2
```



Question 12:

I created a new dataset called Auto_small that contains the first 10 observations and a few selected variables. This makes it easier to work with a smaller portion of the data.
Note: I didn't know name it right, I called it Auto_small and not small_auto

```
> #create a smaller dataset with the first 10 rows and selected columns
> Auto_small <- Auto_clean[1:10, c("mpg", "cylinders", "horsepower", "weight")]
> #display the new dataset
> Auto_small
   mpg cylinders horsepower weight
1   18         8        130   3504
2   15         8        165   3693
3   18         8        150   3436
4   16         8        150   3433
5   17         8        140   3449
6   15         8        198   4341
7   14         8        220   4354
8   14         8        215   4312
9   14         8        225   4425
10  15         8        190   3850
```

.Rhistory × Auto_small ×

Filter

	mpg	cylinders	horsepower	weight
1	18	8	130	3504
2	15	8	165	3693
3	18	8	150	3436
4	16	8	150	3433
5	17	8	140	3449
6	15	8	198	4341
7	14	8	220	4354
8	14	8	215	4312
9	14	8	225	4425
10	15	8	190	3850

Question 13:

I used indexing to view a specific row and column from the dataset. Then, I created a new variable called efficiency that labels cars as high or low efficiency based on their MPG.

```
> # Display the 5th row of the dataset
> Auto_small[5, ]
  mpg cylinders horsepower weight
5  17         8        140   3449
> # Display the mpg column
> Auto_small$mpg
 [1] 18 15 18 16 17 15 14 14 14 15
> # Create a new efficiency column based on mpg
> Auto_small$efficiency <- ifelse(Auto_small$mpg > 25, "high", "low")
> #display new dataset
> Auto_small
   mpg cylinders horsepower weight efficiency
1   18         8        130   3504        low
2   15         8        165   3693        low
3   18         8        150   3436        low
4   16         8        150   3433        low
5   17         8        140   3449        low
6   15         8        198   4341        low
7   14         8        220   4354        low
8   14         8        215   4312        low
9   14         8        225   4425        low
10  15         8        190   3850        low
```

Question 14:

I sorted the dataset by MPG in descending order so that the most fuel-efficient cars appear at the top of the table.

```
> #sorts the dataset from highest to lowest
> Auto_small <- Auto_small [order(-Auto_small$mpg)]
Error in `[.data.frame`(Auto_small, order(-Auto_small$mpg)) :
  undefined columns selected

> Auto_small <- Auto_small [order(-Auto_small$mpg), ]
> #display the sorted dataset
> Auto_small
   mpg cylinders horsepower weight efficiency
1   18         8        130   3504        low
3   18         8        150   3436        low
5   17         8        140   3449        low
4   16         8        150   3433        low
2   15         8        165   3693        low
6   15         8        198   4341        low
10  15         8        190   3850        low
7   14         8        220   4354        low
8   14         8        215   4312        low
9   14         8        225   4425        low
>
```

Question 15:

I used dpyr (After installing it) to complete the task in a single pipeline, but instead of using slice(); I used filter() with row_number() to select the first 10 rows. The remaining steps create a new efficiency variable and sort the data by mpg.

```
> #load the dplyr package
> library(dplyr)
Error in library(dplyr) : there is no package called 'dplyr'

> library(dplyr)
Error in library(dplyr) : there is no package called 'dplyr'

> #I forgot to install it first
> install.package("dplyr")
Error in install.package("dplyr") :
  could not find function "install.package"

> install.packages("dplyr")
WARNING: Rtools is required to build R packages but is not currently installed. Please download and install the appropriate version of Rtools before proceeding:

https://cran.rstudio.com/bin/windows/Rtools/

also installing the dependency 'generics'

trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/generics_0.1.4.zip'
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/dplyr_1.2.0.zip'
package 'generics' successfully unpacked and MD5 sums checked
package 'dplyr' successfully unpacked and MD5 sums checked

The downloaded binary packages are in
        C:\Users\chris\AppData\Local\Temp\RtmpExS4oA\downloaded_packages
> library(dplyr)

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union

> #The %>%, also known as the pipe operator is a feature in the R programming language. It simplifies data manipulation and t
ransformation by allowing users to connect operations seamlessly
> #select(): picks specidic columns
> #filter():subsets row based on conditions
> #arrange():sort the data
> #slice():select specific rows by position
> # %>% (pipe: passes the result of one function to the next)
>
>
> Auto_clean %>%
+ filter(row_number() <= 10) %>%
+ slect(mpg,cylinders, horsepoer,weight) %>%
+ mutate(efficiency = ifelse(mpg >25, "high", "low")) %>%
+ arrange(desc(mpg))
Error in slect(., mpg, cylinders, horsepoer, weight) :
  could not find function "slect"

> Auto_clean %>%
+ filter(row_number() <= 10) %>%
+ select(mpg,cylinders, horsepoer,weight) %>%
+ mutate(efficiency = ifelse(mpg >25, "high", "low")) %>%
+ arrange(desc(mpg))
Error in `select()`:
! Can't select columns that don't exist.
✖ Column `horsepoer` doesn't exist.
Run `rlang::last_trace()` to see where the error occurred.

> Auto_clean %>%
+ filter(row_number() <= 10) %>%
+ select(mpg,cylinders, horsepower,weight) %>%
+ mutate(efficiency = ifelse(mpg >25, "high", "low")) %>%
+ arrange(desc(mpg))
   mpg cylinders horsepower weight efficiency
1   18         8        130   3504        low
2   18         8        150   3436        low
3   17         8        140   3449        low
4   16         8        150   3433        low
5   15         8        165   3693        low
6   15         8        198   4341        low
7   15         8        190   3850        low
8   14         8        220   4354        low
9   14         8        215   4312        low
10  14         8        225   4425        low
>
```

Bonus

```
> Auto_clean %>%
+ filter(mpg > 30) %>%
+ select(mpg, cylinders, weight) %>%
+ head(10)
   mpg cylinders weight
1   31         4   1773
2   35         4   1613
3   31         4   1950
4   32         4   1836
5   31         4   1649
6   32         4   2003
7   31         4   2000
8   33         4   1795
9   33         4   1795
10  32         4   1990
> #Filter() was used to keep cars with mpg >30. Select() was used to slect only the columns mpg, cylinders, and weight. Finally
head() was used to show the first 10 rows of the result
>
```

Using the dplyr concept to do the bonus question, I noticed the filter would be best fit for mpg greater than 30; select() would be best for all of steps to. Finally, I was able to head() to display the results, if I used it with 10 in the parathesis.